STACKS, EXAMPLES OF DEFORMATIONS: LEMMA-POWER-SERIES-RINGS-TI

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1341-1354 of sources/stacks-examples-defos/pinned-original-source.tex; proof span is lines 1356-1411, the complete author proof (54 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 8 of queue_examples-defos.json (pb=2052, stmt 1341-1354, proof 1356-1411).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0DY3: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1072–1079. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In Lemma 0DY3 if $P = k[[x_1, \ldots, x_n]]/(f)$ for some nonzero $f \in (x_1, \ldots, x_n)^2$, then*

(1) *$Inf(\mathcal{D}ef_P)$ is finite dimensional if and only if $n = 1$, and*
(2) *$T\mathcal{D}ef_P$ is finite dimensional if*

$$\sqrt{(f, \partial f/\partial x_1, \ldots, \partial f/\partial x_n)} = (x_1, \ldots, x_n)$$

Proof. Proof of (1). Consider the derivations $\partial/\partial x_i$ of $k[[x_1, \ldots, x_n]]$ over k. Write $f_i = \partial f/\partial x_i$. The derivation

$$\theta = \sum h_i \partial/\partial x_i$$

of $k[[x_1, \ldots, x_n]]$ induces a derivation of $P = k[[x_1, \ldots, x_n]]/(f)$ if and only if $\sum h_i f_i \in (f)$. Moreover, the induced derivation of P is zero if and only if $h_i \in (f)$ for $i = 1, \ldots, n$. Thus we find

$$\mathrm{Ker}((f_1, \ldots, f_n) : P^{\oplus n} \longrightarrow P) \subset \mathrm{Der}_k(P, P)$$

The left hand side is a finite dimensional k-vector space only if $n = 1$; we omit the proof. We also leave it to the reader to see that the right hand side has finite dimension if $n = 1$. This proves (1).

Proof of (2). Let Q be a flat deformation of P over $k[\epsilon]$ as in the proof of Lemma 0DY3. Choose lifts $q_i \in Q$ of the image of x_i in P. Then Q is a complete local ring

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with maximal ideal generated by $q_1, \ldots, q_n$ and ϵ (small argument omitted). Thus we get a surjection

$$k[\epsilon][[x_1, \ldots, x_n]] \longrightarrow Q, \quad x_i \longmapsto q_i$$

Choose an element of the form $f + \epsilon g \in k[\epsilon][[x_1, \ldots, x_n]]$ mapping to zero in Q. Observe that g is well defined modulo (f). Since Q is flat over $k[\epsilon]$ we get

$$Q = k[\epsilon][[x_1, \ldots, x_n]]/(f + \epsilon g)$$

Finally, if we changing the choice of q_i amounts to changing the coordinates x_i into $x_i + \epsilon h_i$ for some $h_i \in k[[x_1, \ldots, x_n]]$. Then $f + \epsilon g$ changes into $f + \epsilon(g + \sum h_i f_i)$ where $f_i = \partial f/\partial x_i$. Thus we see that the isomorphism class of the deformation Q is determined by an element of

$$k[[x_1, \ldots, x_n]]/(f, \partial f/\partial x_1, \ldots, \partial f/\partial x_n)$$

This has finite dimension over k if and only if its support is the closed point of $k[[x_1, \ldots, x_n]]$ if and only if $\sqrt{(f, \partial f/\partial x_1, \ldots, \partial f/\partial x_n)} = (x_1, \ldots, x_n)$. $\square$